

HeatShield

From heatwave alerts to hyperlocal action.

A tool that finds the dangerously hot pockets inside an Indian city, predicts them three days ahead, and then works out where to put cooling relief so it protects the most people.



2
CITIES MAPPED

1,464
GRID CELLS

86 lakh
PEOPLE COVERED

WRITTEN FOR

Anyone picking this project up for the first time — no background in climate science, GIS or web development assumed.

BUILT FOR

PCCOE International Grand Challenge 2026 · Theme: AI for Climate Change · SDG-13.

What you are about to read

Ten pages, in plain English. If you only read one, read this one.

01	Cover
02	Contents & the one-minute version — what this is, in a paragraph
03	The problem — why a citywide heat alert is not enough
04	Who it is for — the people who use it and the people it protects
05	How it works — the whole system in one flowchart
06	How the risk score is built — step by step, with the actual sums
07	The optimizer — the part nobody else builds
08	Tech stack — every piece, and why it is there
09	Compared to what exists — and where we genuinely differ
10	Results, limits, what next — including what we do <i>not</i> claim

THE ONE-MINUTE VERSION

When a heatwave hits an Indian city, the government issues **one warning for the whole city**. But a city is not one temperature. A crowded lane with tin roofs and no trees can be **4–6 °C hotter** than a leafy neighbourhood three kilometres away.

HeatShield splits a city into 1 km squares, scores how dangerous each square is, updates that every day using the live weather forecast, and then answers the question a city official actually has to answer:

“I have five water tankers. Where do I send them?”

Three words that appear on every page

Grid cell

One square of the city, roughly 1 km × 1 km. Pune is cut into 744 of them. Every number in this tool is calculated per cell, not per city.

Vulnerability

How bad heat *would* be here — because of buildings, missing trees and how many people live here. It does not change day to day.

Risk

Vulnerability *plus* today's actual weather. This is the number that changes daily and the one the map shows.

The one sentence to remember: heat *mapping* in India is already solved by several good projects — heat *action* is not. HeatShield is built to close that gap.

WHY THIS NEEDS TO EXIST

One alert for a city that is not one temperature

Heat is one of India's fastest-growing climate killers, and the response has a specific, fixable weakness: it is issued at **city scale**, while the danger exists at **street scale**.

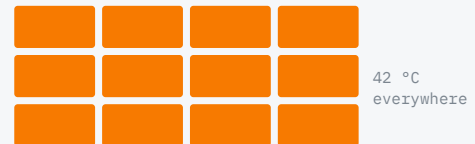
When the weather department forecasts 42 °C for Pune, that is one number for 3.1 million people spread across 700-odd square kilometres. Inside that city:

- A dense lane of tin-roofed houses with no tree cover traps heat and releases it slowly all night. It can run several degrees hotter than the forecast.
- A planned neighbourhood with parks and older trees can run *cooler* than the forecast.
- The hot pockets are usually where the people least able to cope live — outdoor workers, the elderly, families without a fan or a cooler.

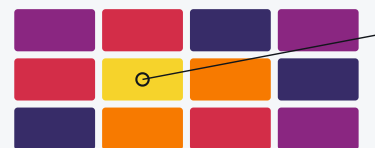
So the alert is technically correct and operationally useless. It tells a ward officer that today is dangerous. It does not tell them **which street**.

FIGURE 1 · THE GAP

What the alert says



What is actually happening



The circled block feels like

48 °C

— six degrees above the number on the news.

Same city, same hour. The warning is uniform; the danger is not.

This is not a theoretical problem

Ahmedabad built South Asia's first Heat Action Plan in 2013 after a deadly 2010 heatwave. It is credited with averting roughly 1,100 deaths a year, and it worked well enough that India's National Disaster Management Authority asked 23 heat-prone states to build their own. Those plans save lives — and they still mostly stop at a citywide alert. The intra-city targeting step was never built.

Why this particular gap is worth attacking

The danger is predictable

Unlike an earthquake, a heatwave is forecast days ahead. The information to act early already exists — it is simply not resolved finely enough to act *locally*.

The hot spots barely move

Buildings and missing trees do not change week to week. The expensive half of the calculation can be done once and reused every day after.

The response is already funded

Cities already own tankers, halls and staff. This is not a request for more money — it is a question of aiming what already exists.

AUDIENCE

Two groups: who uses it, and who it protects

GROUP 1 · THE USERS

People who decide where resources go

They open the dashboard. They are not data scientists — they are administrators with a budget, a map and a bad week ahead.

- **Municipal disaster management cells** — the office that actually deploys tankers, opens relief centres and issues local warnings.
- **City health departments** — who need to know which health posts to stock and which neighbourhoods to send ASHA workers into.
- **Ward officers** — the person on the ground with four tankers and two community halls, deciding on a Monday morning.
- **Urban planners** — who can see, over time, which wards keep appearing in the danger band and need trees or shade structures.

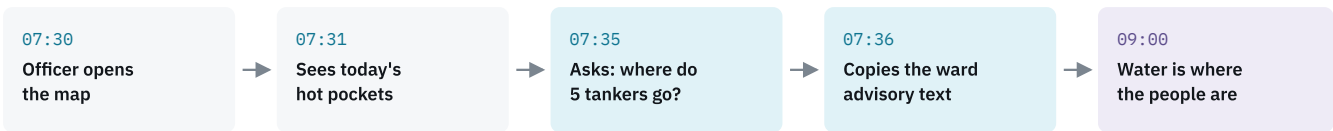
GROUP 2 · THE BENEFICIARIES

People who never open the app

They benefit because relief arrives closer to them. They are the reason the scoring weights population the way it does.

- **Outdoor workers** — construction labour, street vendors, delivery riders, traffic police. Working through the hottest hours is not optional for them.
- **Elderly residents**, especially those living alone, whose bodies regulate temperature poorly and who may not seek help.
- **Infants and pregnant women**, both at elevated medical risk in extreme heat.
- **Households without cooling** — where a fan is the only defence and a power cut removes even that.

A day in the life — what actually happens



Total time from opening the tool to having a defensible deployment plan: under ten minutes.
No GIS training, no spreadsheet, no data team.

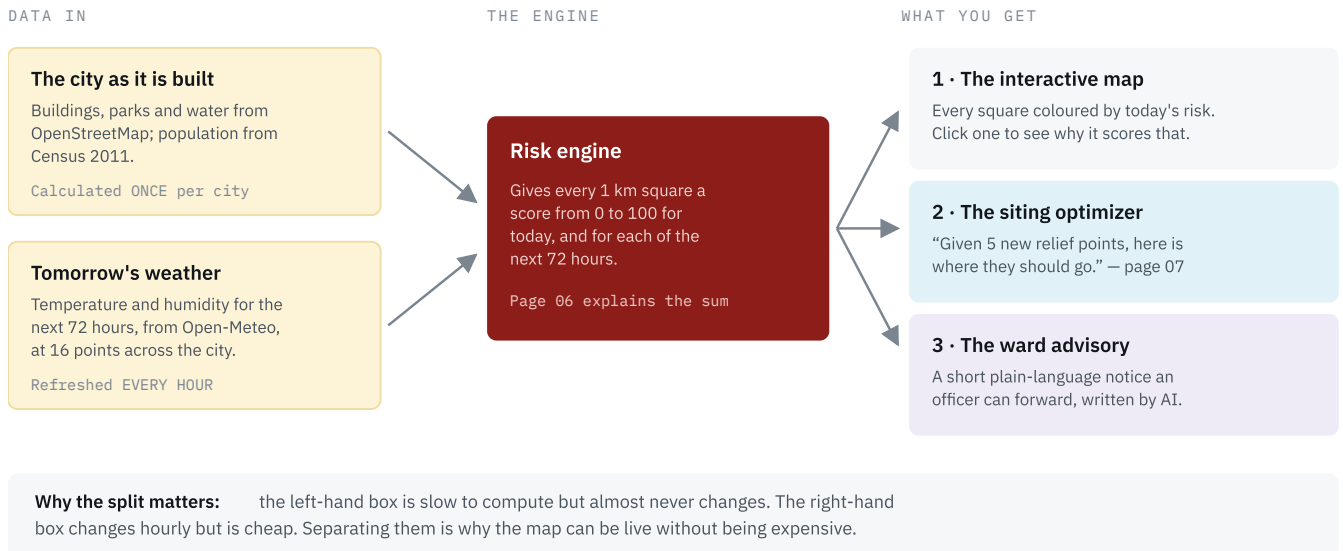
What each group needs, and where the tool answers it

WHO	WHAT THEY NEED TO KNOW	WHERE THE TOOL ANSWERS IT
Ward officer	“Which streets in my ward are worst today?”	The map, zoomed to their ward — click any square for its breakdown.
Disaster management cell	“Where do the tankers and relief centres go?”	The optimizer panel — a ranked list of placements and the people each one reaches.
Health department	“Which health posts should be stocked, and when?”	The 72-hour trend chart — shows which day the worst square crosses the relief threshold.

SYSTEM ARCHITECTURE

The whole system, in one picture

Read it left to right. Two kinds of information come in on the left, they are combined in the middle, and three useful things come out on the right.



What happens when you click a square

The map never asks you to trust a number you cannot open up. Clicking any square shows the whole chain that produced its score:

- The **risk band** in plain words — “Severe: relief centres should open, restrict outdoor work 12:00–16:00.”
- How many **people live there**, and what it **feels like** versus the regional forecast.
- The three ingredients — built-up density, green cover, distance to water — each as a bar, so you can see *why* this square scores what it does.
- A button to draft the ward advisory for that exact neighbourhood.

Why it stays fast

The static layer is read from memory rather than fetched per request, and the forecast is cached for an hour. Dragging the time slider recomputes 744 squares and redraws the map in a few hundred milliseconds.

If the weather service is slow or down, the app serves the last good forecast and labels it “cached” rather than showing a spinner or, worse, a blank map.

Once

THE CITY LAYER IS COMPUTED PER CITY, THEN REUSED

Hourly

THE FORECAST REFRESHES ON ITS OWN

Instantly

THE MAP RECOMPUTES AS YOU MOVE THE SLIDER

THE MATHS, WITHOUT THE JARGON

How one square gets its score

Step 1 — What is built here (done once)

For each 1 km square we count three things from free public data:

Buildings	How many building footprints are inside the square. More concrete means more heat stored and released at night.
Green cover	How much of the square is park, forest or grass. Trees cool by shading and evaporation.
Water nearby	How close the nearest lake or river is. Water cools its surroundings, but only for about a kilometre.

These are blended into a single **heat proxy**, then every square is ranked against every other square in the same city.

Step 2 — Who is here

The city's Census population is spread across the squares in proportion to how many buildings each one has. A square with 5% of the city's buildings gets about 5% of its people. This is a standard technique called *dasymetric mapping*.

Heat proxy and population density are combined into **vulnerability**, scored 0–100. This never changes day to day.

Step 3 — Add today's weather

Three sub-steps, and this is where the number becomes live:

- The forecast is a regional number, so each square gets an **urban heat island bump** of up to +4 °C, scaled by how built-up it is. That gives a “feels like here” temperature.
- That temperature is scored against the internationally used **heat index danger bands** — below 27 °C is no risk, 41 °C is “danger”, 54 °C is the top of the published scale.
- The result is then weighted by vulnerability, because the same 45 °C is a worse emergency in a crowded block than on an empty ridge.

FIGURE 3 · A WORKED EXAMPLE

Regional forecast	37.7 °C
+ heat island bump	+3.1 °C
Feels like here	40.8 °C
Heat-index stress	53 / 100
Vulnerability here	79 / 100
Final risk score	64 — Severe

Real output from a cell in Pune. Every one of these numbers is shown in the app when you click a square — the tool never asks you to trust a score you cannot open up.

Why ranking, not raw values

Our first attempt scaled scores between the lowest and highest building count. Because a few squares have enormous counts, that pushed the *median* square to 76/100 — the whole city painted “severe”, which is exactly the failure this project exists to fix. Ranking each square against the others instead gives a proper spread: in Pune the 10th, 50th and 90th percentile squares now score 12, 50 and 91.

A colour choice that matters



The scale gets steadily *brighter* as risk rises, not just redder. That means it still works for red–green colourblind readers and in black-and-white printouts — which the usual green-to-red heat map does not.

THE DIFFERENTIATING FEATURE

Where should the water go?

Every other part of this project exists to make this question answerable. This is the piece comparable tools do not attempt.

The idea in one paragraph

You have a city full of squares, each with a risk score and a population. You already have some relief — public water points. And you have a budget: say five new places you could set one up. The optimizer tries to pick the five that bring the **most currently-unreached at-risk people** within a short walk of relief.

Why “greedy”?

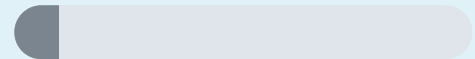
The algorithm takes the single best option available at each step, then re-checks. It does not try every combination — with 744 possible spots that would be billions of comparisons.

For coverage problems of this shape that shortcut is mathematically well understood: it is guaranteed to land within about **63% of the theoretically perfect answer**, and it runs in roughly **40 milliseconds**. That speed is what lets the map redraw the instant you drag the budget slider, which is what makes it feel like a tool rather than a report.

It is also explainable one site at a time — “this one reaches 41,000 people nobody currently reaches” — which matters when someone has to justify the decision.

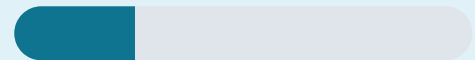
FIGURE 4 · PUNE, HEATWAVE SCENARIO

Reached by relief that exists today



9.4% · 1,17,927 people

With five optimally placed new points



26.2% · 3,28,098 people

+2,10,171

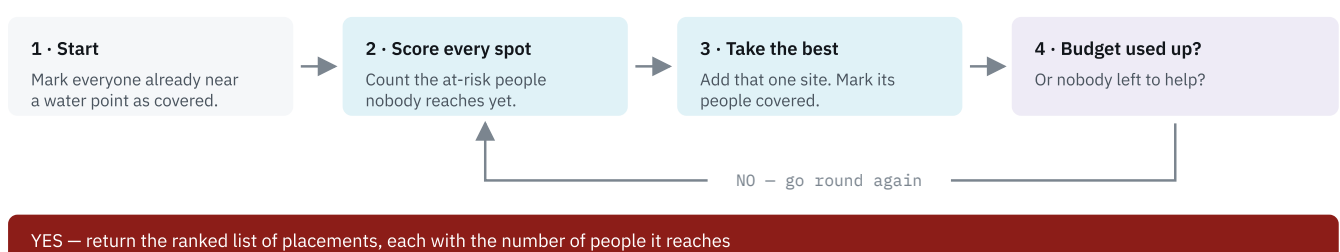
MORE PEOPLE REACHED, FROM FIVE PLACEMENTS

One honest assumption, made visible

OpenStreetMap lists 714 hospitals and 450 clinics in Pune. If we counted all of them as “heat relief”, existing coverage jumps to 94% and the whole question looks solved.

But a private clinic is not a cooling shelter, and a hospital is where you go *after* heatstroke, not to avoid it. So the default counts only the **26 public water points** Pune has mapped for 3.1 million people — and the dashboard exposes this as a switch, so anyone can disagree with us and see the other answer.

How it decides, step by step



WHAT IT IS BUILT FROM

Every piece, and why it is there

Everything below is free. There is no paid service anywhere in this project — deliberately, so a municipality could run it without a procurement process.

PIECE	WHAT IT IS	WHY WE CHOSE IT
Next.js + TypeScript	The website and the server, in one	One language front to back. TypeScript catches mistakes before they reach a user — useful when numbers on screen represent danger.
MapLibre GL	The interactive map	Free and open source, no API key, no per-view billing. Draws 744 coloured squares smoothly on a phone.
OpenFreeMap	The street map underneath	Free map tiles with no key and no usage cap. We wrote a custom dark style so the streets stay readable under the heat colours.
Neon (PostgreSQL)	The database	Stores the grid, facilities, cached forecasts and advisories. Free tier, and it wakes from sleep fast enough for a live demo.
Drizzle ORM	Talks to the database	Lightweight and typed, so a wrong column name is a compile error rather than a 3 a.m. bug.
Open-Meteo	The weather forecast	Free, no signup, good India coverage. We are explicit that this is <i>not</i> IMD — IMD has no stable public API, and claiming otherwise would be dishonest.
OpenStreetMap + Overpass	Buildings, parks, hospitals	The only openly licensed source detailed enough to see individual buildings in Indian cities. 246,116 buildings pulled for Pune alone.
Census of India 2011	Population totals	One official number per city, spread across squares by building count.
Google Gemini	Writes the ward advisories	Free tier. Given only the square's real numbers and told to invent nothing — a made-up temperature in a public safety notice is worse than no notice.
React Three Fiber + Recharts	The 3D surface and the trend chart	The chart plots the worst square against the median one, because the gap between those two lines is the whole argument. The 3D surface is used once, as illustration.
Vercel	Where it runs	Free hosting that deploys straight from GitHub.

Written by hand, on purpose

The optimizer is our own TypeScript, not a library and not a separate Python service. It is a small, well-understood algorithm; adding a second language to the deployment would have bought nothing and cost a lot of fragility.

Checks that run on every build

The risk model and the optimizer both ship with assert-based self-checks — including one that fails the build if the colour ramp ever stops getting brighter as risk rises, because that is what keeps it colourblind-safe.

HONEST POSITIONING

What already exists, and where we actually differ

We are not claiming to have invented heat mapping. Several Indian projects already do it, some better than we do.

APPROACH	HYPERLOCAL MAPPING	UPDATES DAILY	SAYS WHERE TO PUT RELIEF	WHAT IT IS
IMD heat alerts	No	Yes	No	The official warning system. Accurate, authoritative, and issued for a whole city or district.
Ahmedabad Heat Action Plan	Partly	Yes	Partly	The gold standard for response — alert days, hospital readiness, public messaging. Planning is citywide; siting is decided by officials, not computed.
SEEDS + Microsoft “Sunny Lives”	Yes	No	No	Excellent building-level AI risk mapping and household warnings. Proves hyperlocal works, then stops at identifying risk.
IIT Mandi building-level heat maps	Yes	No	No	Rigorous academic mapping at building resolution. A research output, not an operational tool.
HeatShield	Yes	Yes	Yes	1 km grid on a live 72-hour forecast, plus a coverage optimizer returning ranked placements and a sendable advisory.

The three things that are genuinely ours

01

The action layer

Existing work ends at a map. We take the map as the *input* and produce a decision: coordinates, ranked, with the population each reaches.

02

Live, not an atlas

Most hyperlocal heat maps are historical — “where is it usually hot”. Ours answers “where is it dangerous *this Friday afternoon*”.

03

Built for the last mile

The output is not a dashboard for analysts. It is a paragraph an officer can paste into WhatsApp, naming the real neighbourhood, with three actions for today.

Where we are weaker, and we should say so: SEEDS and IIT Mandi resolve heat at *building* level using real satellite land-surface temperature. We work at 1 km squares using a proxy. Their science is better; our contribution is the step after the map. A better input layer would make our optimizer better, not redundant.

WHERE THE PROJECT STANDS

What works, what does not, and what comes next

Built and working today

1,464GRID CELLS ACROSS PUNE
AND AHMEDABAD**246,116**BUILDING FOOTPRINTS
PROCESSED FOR PUNE**~40 ms**TO RECOMPUTE THE
OPTIMIZER LIVE

- A working dashboard with a live, clickable ward-level risk map for two Indian cities.
- A 72-hour forecast fused into the risk surface, refreshed hourly, with a trend chart.
- A siting optimizer with a before-and-after coverage comparison.
- AI-drafted advisories, named to the real neighbourhood, cached per area per day.

What we deliberately do not claim

- **This is a heat proxy, not satellite temperature.** Real land-surface temperature needs Google Earth Engine, which needs an approval we chose not to wait on. Every component is stored separately so real data can replace the proxy without rewriting anything.
- **Scores are relative to their own city.** A 70 in Pune and a 70 in Ahmedabad are not the same absolute danger. Absolute danger enters through the forecast, not the vulnerability layer.
- **The forecast is Open-Meteo, not IMD.** The interface credits it by name rather than implying an official source.
- **Population is estimated, not counted.** Census totals spread by building count is a standard method, but it is an estimate.
- **The advisories are machine-drafted.** Every one is labelled as such and carries a read-before-sending notice.

Where it goes next

More cities

The pipeline is city-agnostic. Adding a city means running the seed script once — no code changes.

Better inputs

Swap the heat proxy for real satellite land-surface temperature, and ward boundaries for the 1 km grid, and every number gets sharper.

Beyond heat

Map → forecast → optimise → advise is a general pattern. Flooding and air quality fit it without redesign.

In one line: India already knows how to draw a good heat map. HeatShield is an argument that the valuable, unbuilt part is what you do with it the morning after.